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Poster #111 - Machine Learning Reveals Welfare Discourse Embedded in Japanese Sentencing: NLP Analysis of 2,002 Theft Judgments

In Event: *Poster Session I (posters 80 – 155)*

Thu, Nov 19, 6:30 to 7:30pm, Monroe Room - Posters 80 - 155 - 6th Floor

Abstract

Introduction. This study uses NLP and machine learning to examine whether welfare-support vocabulary is structurally embedded in Japanese criminal sentencing, across 2,002 first-instance theft judgments.

Design/Methods. Judgments from Westlaw Japan (2010-2023) were vectorized via TF-IDF (5,000 features). Two interpretable models were trained: L1-regularized logistic regression with exact SHAP values, and fine-tuned Japanese BERT with three-fold Integrated Gradients (IG) attribution. A pre-specified 59-term welfare lexicon, organized into six semantic subcategories, was evaluated against importance rankings.

Results. Linear AUC = 0.913; BERT AUC = 0.951 (+3.7 points, concentrated in welfare-dense judgments, Mann-Whitney p = 0.007). Of 59 welfare terms, 14 fell in SHAP top 10% (2.4x chance). Three-fold IG corroborates at the BERT representation level: welfare tokens receive 2.23x (range 2.05-2.50x) the mean IG magnitude of non-welfare tokens ($p < 10^{-20}$ per fold), dominating the top-500 ranks at 5.21x chance (range 5.14-5.33x). The 24 suspension / 35 custody split is category-structured: abstract social-rehabilitation frames drive suspension, while concrete family-guarantor frames lean toward custody.

Discussion. As the first NLP/ML study of Japanese criminal sentencing, this shows judges use welfare concepts as structured discursive resources—not peripheral vocabulary. The subcategory-resolved findings ground the PSI reform debate, specifying welfare-language domains a formal system should systematize.

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